

Summary

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- The *average rate of change*, $AvRCh$, of a function f across an interval $[a, b]$ is given by

$$\begin{aligned} AvRCh &= \frac{\Delta f}{\Delta x} \\ &= \frac{f(b) - f(a)}{b - a} \end{aligned}$$

- The slope, m_{sec} , of the line secant to the graph of a function f at the points $(a, f(a))$ and $(b, f(b))$ is given by

$$\begin{aligned} m_{sec} &= \frac{\Delta y}{\Delta x} \\ &= \frac{f(b) - f(a)}{b - a} \end{aligned}$$

- The *derivative* of the function f at a , denoted $f'(a)$, is given by

$$f'(a) = \lim_{x \rightarrow a} \frac{f(x) - f(a)}{x - a} = \lim_{h \rightarrow 0} \frac{f(a + h) - f(a)}{h}$$

and represents the instantaneous rate of change of f with respect to x at $x = a$ and also the slope of the line tangent to the graph of $y = f(x)$ at the point $(a, f(a))$.